

counterparts in other mammals. This approach has revealed a number of genes that are apparently changing (evolving) faster in the human than in either the chimpanzee or the mouse. Among them are genes involved in defense against malaria and tuberculosis as well as at least one gene that regulates brain size. When genes are classified by function, the genes that seem to be evolving the fastest are those that code for transcription factors. This discovery makes sense because transcription factors regulate gene expression and thus play a key role in orchestrating the overall genetic program.

One transcription factor of interest, whose gene is called *FOXP2*, might be involved with the acquisition of speech in humans. Attention was first drawn to *FOXP2* in 2002 when a research publication reported that this gene was evolving faster in humans (*Homo sapiens*) than in other primates, based in part on two amino acids found only in the human protein sequence. The researchers proposed that this was a human-specific version of the gene that resulted from strong natural selection and might have conferred on humans the ability to speak.

In 2014, a high-quality sequence of the Neanderthal (*Homo neanderthalensis*) genome was achieved from a small amount of preserved genomic DNA, and shortly thereafter the DNA of another member of the species *Homo*, the Denisovan, was sequenced as well (see Concept 34.7). It turns out that some DNA sequences from both species encode the same two amino acids, so the two amino acids are not human-specific. In a paper published in 2018, researchers analyzed genomic sequences from a larger and more diverse population of modern humans. In this more robust study, the researchers found no evidence for strong natural selection for those two amino acids in the human lineage during the time frame relevant to language acquisition.

However, the *FOXP2* gene is still interesting in its own right. Several lines of evidence suggest that the *FOXP2* gene product regulates genes that function in vocalization in vertebrates. First, mutations in this gene can produce severe speech and language impairment in humans. Second, the *FOXP2* gene is expressed in the brains of zebra finches and canaries at the time when these songbirds are learning their songs. And third, perhaps the strongest evidence comes from a “knockout” experiment in which researchers disrupted the *FOXP2* gene in mice and analyzed the resulting phenotype (**Figure 21.18**). The homozygous mutant mice had malformed brains and failed to emit normal ultrasonic vocalizations; mice with one faulty copy of the gene also showed significant problems with vocalization. These results support the idea that the *FOXP2* gene product turns on genes involved in vocalization.

Expanding on this analysis, another research group replaced the *FOXP2* gene in mice with a “humanized” copy coding for the human versions of the *FOXP2* protein, with the two amino acids that differ between human and chimpanzee sequences. The researchers reported that although the mice with the human *FOXP2* gene were generally healthy,

they had subtly different vocalizations and showed changes in brain cells in circuits associated with speech in human brains. While the *FOXP2* gene clearly affects vocalization and speech, the precise mechanism by which it does this is still to be determined.

The *FOXP2* story is an excellent example of how different approaches can complement each other in uncovering biological phenomena of widespread importance. The *FOXP2* experiments used mice as a model for humans because it would be unethical (as well as impractical) to carry out such experiments in humans. Mice and humans, which diverged about 65.5 million years ago (see Figure 21.17), share about 85% of their genes. This genetic similarity can be exploited in studying human genetic disorders. If researchers know the organ or tissue that is affected by a particular genetic disorder, they can look for genes that are expressed in these locations in mice.

Even though more distantly related to humans, fruit flies have also been a useful model species for study of such human disorders as Parkinson’s disease and alcoholism, while nematodes (soil worms) have yielded a wealth of information about aging. Further research efforts are under way to extend genomic studies to many more species, including neglected species from diverse branches of the tree of life. These studies will advance our understanding of evolution, of course, as well as all aspects of biology, from human health to ecology.

Comparing Genomes Within a Species

Another exciting consequence of our ability to analyze genomes is our growing understanding of the spectrum of genetic variation in humans. Because the history of the human species is so short—probably about 200,000 years—the amount of DNA variation among humans is small compared to that of many other species. Much of our diversity seems to be in the form of single nucleotide polymorphisms (SNPs). SNPs are single base-pair sites where genetic variation is found in at least 1% of the population (see Concept 20.2); they are usually detected by DNA sequencing. In the human genome, SNPs occur on average about once in 100–300 base pairs. Scientists have already identified the locations of several million SNP sites in the human genome and continue to find additional SNPs. These are stored in databases around the world, one of which is run by the National Center for Biotechnology Information (NCBI) and can be accessed at <http://www.ncbi.nlm.nih.gov/SNP/>.

In the course of this search, they have also found other variations—including chromosomal regions with inversions, deletions, and duplications. The most surprising discovery has been the widespread occurrence of *copy-number variants* (CNVs), loci where some individuals have one or multiple copies of a particular gene or genetic region rather than the standard two copies (one on each homolog). CNVs result from regions of the genome being duplicated or deleted inconsistently within the population. One study of 40 people found more than 8,000 CNVs involving 13% of the genes in